Database Mail Checks

Database Mail is a .NET application and uses the latest TLS version for sending emails (TLS 1.2 is the current version at the time of this document writing) so you need to confirm on the server itself that it needed .NET 4.8 (or later) and proper TLS 1.2 Crypo keys within the registry in order for it to work.

From within the Operating System that hosts the SQL server instance your setting Database Mail up on. Open a PowerShell termi nal as Administrator, and run the following commands...

```
reg query "HKLM\SOFTWARE\Microsoft\NET Framework Setup\NDP\v4\Full" /v Release

[Net.ServicePointManager]::SecurityProtocol

reg query "HKLM\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Client"
reg query "HKLM\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Server"
reg query "HKLM\SOFTWARE\Microsoft\.NETFramework\v4.0.30319" /v SchUseStrongCrypto
reg query "HKLM\SOFTWARE\WOW6432Node\Microsoft\.NETFramework\v4.0.30319" /v SchUseStrongCrypto
```

Expected Output:
- .NET version should resolve to 4.8 or higher
- TLS protocols should include `Tls12`
- Registry values should return expected keys and values

The first command determines what .NET version you have installed. It will give you a hex based value such as " **0x80eb1**" this indicates .NET 4.8 is installed. You can look up your returned .NET code using the table below. If it's using a version prior to 4.8 you will need to update .NET to 4 .8 or later.

.NET 4.5 - 4.8.1 Registry Hex Codes

.NET Version	Release Key (Hex)	Decimal Value
.NET 4.5	**0x5C07D**	379893
.NET 4.5.1	**0x60047**	393295
.NET 4.5.2	**0x60049**	393297
.NET 4.6	**0x7107E**	460798
.NET 4.6.1	**0x6041F**	394271
.NET 4.6.2	**0x60636**	394806
.NET 4.7	**0x70805**	460805
.NET 4.7.1	**0x709FE**	461310
.NET 4.7.2	**0x70BF6**	461814
.NET 4.8	**0x80EB1**	528049
.NET 4.8.1	**0x81E58**	533320

The second command is to list the TLS versions that are supported, it should list "**Tls12**".

The remaining commands are to ensure you have the proper Crypto Registry Keys. They should return the values above, if not th ose will need to be updated/added using the commands below

```
New-Item -Path "HKLM:\SOFTWARE\Microsoft\.NETFramework\v4.0.30319" -Force | Out-Null
New-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\.NETFramework\v4.0.30319" -Name "SchUseStrongCrypto" -Value 1 -PropertyType DWord -Force | Out-Null
New-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\.NETFramework\v4.0.30319" -Name "SystemDefaultTlsVersions" -Value 1 -PropertyType DWord -Force | Out-Null

New-Item -Path "HKLM:\SOFTWARE\WOW6432Node\Microsoft\.NETFramework\v4.0.30319" -Force | Out-Null
New-ItemProperty -Path "HKLM:\SOFTWARE\WOW6432Node\Microsoft\.NETFramework\v4.0.30319" -Name "SchUseStrongCrypto" -Value 1 -PropertyType DWord -Force | Out-Null
New-ItemProperty -Path "HKLM:\SOFTWARE\WOW6432Node\Microsoft\.NETFramework\v4.0.30319" -Name "SystemDefaultTlsVersions" -Value 1 -PropertyType DWord -Force | Out-Null
```

Additionally you may need to add the SCHANNEL keys as well...

```
New-Item -Path "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Client" -Force | Out-Null
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Client" `
  -Name "Enabled" -Value 1 -PropertyType DWord -Force | Out-Null
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Client" `
```

```
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Client" `
  -Name "DisabledByDefault" -Value 0 -PropertyType DWord -Force | Out-Null

New-Item -Path "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Server" -Force | Out-Null
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Server" `
  -Name "Enabled" -Value 1 -PropertyType DWord -Force | Out-Null
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.2\Server" `
  -Name "DisabledByDefault" -Value 0 -PropertyType DWord -Force | Out-Null
```

Once these are added you will need to reboot the server in order for changes to take effect. After that then continue to test Database Mail again to ensure it is working.